

LEARNING EEG REPRESENTATIONS FOR SLEEP DEPRIVATION CLASSIFICATION AND SLEEPINESS PREDICTION

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1 PROBLEM STATEMENT

This project aims to develop a deep learning framework for analyzing resting-state EEG under normal sleep (NS) and sleep deprivation (SD) conditions. Our primary objective is to train a CNN-based classifier to determine whether a given EEG epoch belongs to the NS or SD condition, allowing us to learn discriminative representations of sleep-loss-related neural activity. Beyond this main classification task, we will investigate whether these learned representations also encode meaningful behavioral information associated with sleep deprivation. In particular, we will focus on two downstream targets: the Karolinska Sleepiness Scale (KSS) and the Stanford Sleepiness Scale (SSS), both of which are standard subjective measures of momentary sleepiness and alertness, and are therefore closely related to the physiological effects of sleep deprivation. To model these downstream targets, we will extract intermediate feature representations from the trained CNN for each EEG epoch, aggregate these representations across all epochs belonging to the same recording session to obtain a session-level embedding, and then train a second deep learning model on top of these aggregated embeddings to predict KSS and SSS scores. In this way, the project examines not only whether resting-state EEG can be used to identify sleep deprivation, but also whether the representations learned for this task capture the severity of its immediate behavioral effects.

2 WHY IS THIS PROBLEM INTERESTING?

This project is interesting because it attempts to bridge the gap between the brain’s physiological response to sleep deprivation and an individual’s subjective experience of it. EEG signals objectively capture the brain’s response to sleep deprivation, while sleepiness scales like KSS and SSS show how a person actually feels, and the two do not always align neatly. The central question we aim to address is whether a model trained purely on brain signals can predict something as personal as the subjective feeling of sleepiness. This matters because it tests the explanatory power of the deep learning features. This has broader implications in neurotechnology, where the objective EEG signals can one day serve as a reliable alternative to the self-reported fatigue measures in healthcare, aviation, and long-haul driving.

3 DATASET

The project will use the Resting-State EEG Dataset for Sleep Deprivation introduced in Xiang et al. (2024), which contains EEG recordings from 71 participants collected under two within-subject conditions: normal sleep (NS) and sleep deprivation (SD). In each session, participants underwent 5

minutes of resting-state EEG recording with 61 electrodes sampled at 500 Hz, along with behavioral assessments related to alertness, mood, and sleepiness. For our work, we focus on the eyes-open resting-state EEG recordings and use the associated Karolinska Sleepiness Scale (KSS) and Stanford Sleepiness Scale (SSS) as downstream behavioral targets. The paired NS–SD design makes this dataset well suited for studying how sleep loss alters neural activity while also enabling us to examine whether learned EEG representations capture subjective sleepiness in addition to condition-level differences.

4 RELATED WORK

Recent work on this sleep deprivation EEG dataset has mainly followed two directions: statistical EEG analysis and direct condition classification. The dataset paper (Xiang et al., 2024), introduced the benchmark and validated it through spectral and behavioral comparisons between normal sleep and sleep deprivation. Follow-up studies Bai et al. (2024); Liu et al. (2025) analyzed handcrafted EEG descriptors, including spectral parameters and phase-amplitude coupling, and related them to vigilance or subjective sleepiness. These studies provided physiological insight, but did not study learned neural representations.

A more recent predictive study (Kumar et al., 2025) compared feature-based machine learning models with CNNs, LSTMs, and Transformers for normal-sleep versus sleep-deprived classification. However, the study did not consider the prediction of SSS and KSS metrics and its focus was limited the performance on direct classification using standard architectures. Finally, Del Pup et al. (2025) included this dataset in a broader benchmark of EEG deep learning pipelines and showed that preprocessing choices substantially affect performance, with minimally filtered data often outperforming raw or heavily processed signals.

5 GOALS

The primary goal of this project is to determine whether resting-state EEG contains sufficient information to distinguish between NS and SD with the use of a CNN-based classifier. As a secondary objective, we aim to test whether the representations learned by this classifier also capture behaviorally meaningful information related to subjective sleepiness. Through this two-stage setup, we aim to study both condition-level discrimination and the extent to which learned EEG features transfer to downstream behavioral prediction. First, we train the CNN at the epoch level for NS versus SD classification, and then extract intermediate feature representations for each epoch from the trained network. These representations will be aggregated across epochs within a recording session to obtain a session-level embedding, which will be used to train a separate deep learning model for predicting KSS and SSS scores. We will evaluate the model on both the primary and downstream tasks. For the NS and SD classification task, we will report classification performance accuracy and F1-score. For the downstream tasks, we will first model KSS and SSS as continuous targets and evaluate prediction quality using regression-based metrics.

6 PLAN AND MILESTONES

- **Week 1:** Understand the dataset structure, preprocess the EEG recordings, define the epoching strategy, and finalize the train/validation/test split.
- **Week 2:** Implement and train a baseline CNN for the primary NS vs. SD classification task.
- **Week 3:** Refine the classifier, extract intermediate epoch-level features, and build the aggregation pipeline to obtain session-level embeddings.
- **Week 4:** Train and evaluate the downstream model for predicting KSS and SSS, first as continuous targets and, if needed, as binary high-vs.-low sleepiness labels.
- **Week 5:** Analyze results, prepare tables and figures, and finalize the report and presentation.

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